Hoyoung Kim

khy@dankook.ac.kr | COIS Lab

Education

Dankook University — Cheonan, Republic of Korea
M.S. in Industrial Engineering — *Mar. 2026 – Present*

- Computational Optimization for Intelligent Systems (COIS) Lab

Dankook University — Cheonan, Republic of Korea
B.S. in Management Engineering GPA: 4.09/4.50 — *Graduated Feb. 2026*

Experience

Dankook University — Cheonan, Republic of Korea
Researcher — *Dec. 2024 – Present*

- Conducted public- and private-sector research projects on data-driven prediction, mathematical optimization, and decision making in industrial systems at the COIS Lab.

Skills

Programming Languages: Python, SQL

Optimization: Gurobi

Technical Software: AutoMod, AutoCAD

Certifications: SQL Developer (SQLD)

Publications(domestic)

1. Han, G., Noh, Y., Woo, M., Hong, J., **Kim, H.**, Kim, H., Cho, S. W.† (2026). Social Network Analysis of Regional Health Recognition Inequality Factors: A Focus on Secondhand Smoke. *Health and Social Welfare Review.* (KCI)
2. **Kim, H.**, Kang, S., Seol, H., Yoo, C., Hong, B., Cho, S. W.† (2025). Patent-Driven Topic Modeling and Opportunity Analysis for Smart Logistics Technology. *Journal of Logistics Science & Technology.* (KCI)

Conferences(international)

1. **Kim, H.**, Kim, J., Woo, M., Cha, Y., Zhou, Y., Lee, W.†, Cho, S. W.† (2026). Time-Dependent Ship Speed Optimization with Physics-Informed Fuel Oil Consumption Prediction. *GRN-BRI 2026*, Jeonju, Republic of Korea. Nov. 2026.
2. Woo, M., Kim, H., **Kim, H.**, Yoo, T., Cho, S. W.† (2026). TDMA Slot Assignment for Efficient VHF Data Exchange in Maritime Wireless Networks. *INFORMS TNA 2026*, Online. Oct. 2026. [**Best Student Paper Award**]
3. Woo, M., Kim, H., **Kim, H.**, Zhou, Y., Cho, S. W.† (2026). Neural Column Generation for Quay Crane Scheduling. *International Conference on Logistics and Maritime Systems (LOGMS 2026)*, Barcelona, Spain. Jul. 2026. [**Best Paper Award**]

Conferences(domestic)

1. **Kim, H.**, Han, G., Kim, J., Woo, M., Kim, H., Lee, W., Cho, S. W.† (2026). 머신러닝 기반의 연료 소모량 예측을 활용한 선박 속도 최적화. *KASOTS 2026*, Jeju, Republic of Korea. May 2026.
2. **Kim, H.**, Kang, S., Seol, H., Yoo, C., Hong, B., Cho, S. W.† (2025). 스마트 물류 기술 동향 분석을 위한 특허 데이터 기반 토픽모델링 및 기술 기회 분석. *LOGIST 2025*, Daegu, Republic of Korea. Nov. 2025.

Projects

AI Job Analysis & Workforce Impact Assessment Aug. 2026 – Oct. 2026
Korean Federation of Community Credit Cooperatives

- Extracted action verbs and object terms from task descriptions using natural language processing; mapped them to O*NET tasks to quantify task-level AI exposure; and established criteria to classify AI impact as automation, augmentation, or minimal impact by job function.

Data-Driven FOC Prediction & Speed Optimization May 2026 – Nov. 2026
Korea Research Institute of Ships & Ocean Engineering (KRISO)

- Developed an ML-based fuel-oil consumption (FOC) prediction module reflecting real-world maritime environments, and formulated a voyage-wide speed optimization model to minimize total fuel usage.

Credit Rating Model Development Sep. 2025 – Dec. 2025
NICE Information Service

- Automated corporate credit rating models using financial and non-financial data and multivariate statistical analysis, and developed a specialized evaluation framework for the shipping industry.

AI Investment Model & Capability Modeling Jul. 2025 – Dec. 2025
Korea Technology Finance Corporation (KOTEC)

- Analyzed corporate innovation capabilities from linear and nonlinear perspectives, and developed leading indicators of future innovation and growth potential applicable to early-stage startups.

VDES Network Resource Allocation Optimization Feb. 2025 – Dec. 2027
Ministry of Oceans and Fisheries

- Formulated an optimization model for efficient allocation of base-station communication resources in VHF Data Exchange Systems (VDES), supporting data exchange among coastal stations, vessels, and satellites.

Dynamic Cell Planning (DCP) Algorithm Development Sep. 2025 – Nov. 2025
AIVeNautics

- Formulated an optimization model considering satellite-based internet services such as Starlink, ship-specific broadcasting ranges, traffic characteristics, and channel usage patterns.

Timetable Optimization Service (TOS) Algorithm Jul. 2025
Dankook University

- Developed a user-preference-based timetable optimization service for undergraduate course planning.

Awards and Honors

Excellence Award, LOGIST 2026 Logistics Paper Competition Nov. 2026

Best Student Paper Award, 17th INFORMS Telecommunications and Network Analytics Conference (TNA 2026) Oct. 2026

Best Paper Award, 14th International Conference on Logistics and Maritime Systems (LOGMS 2026) Jul. 2026

Top Excellence Award, LOGIST 2025 Logistics Paper Competition Nov. 2025

Encouragement Award, 21st KMAC Management Innovation Competition Nov. 2025

Beomeun Scholarship Foundation, Scholarship Student Nov. 2025

Excellence Award, DKU CTL Best Practice Competition Jul. 2025

Academic Excellence Awards, Dankook University (성적우수상, 우등상) 2024 – 2026